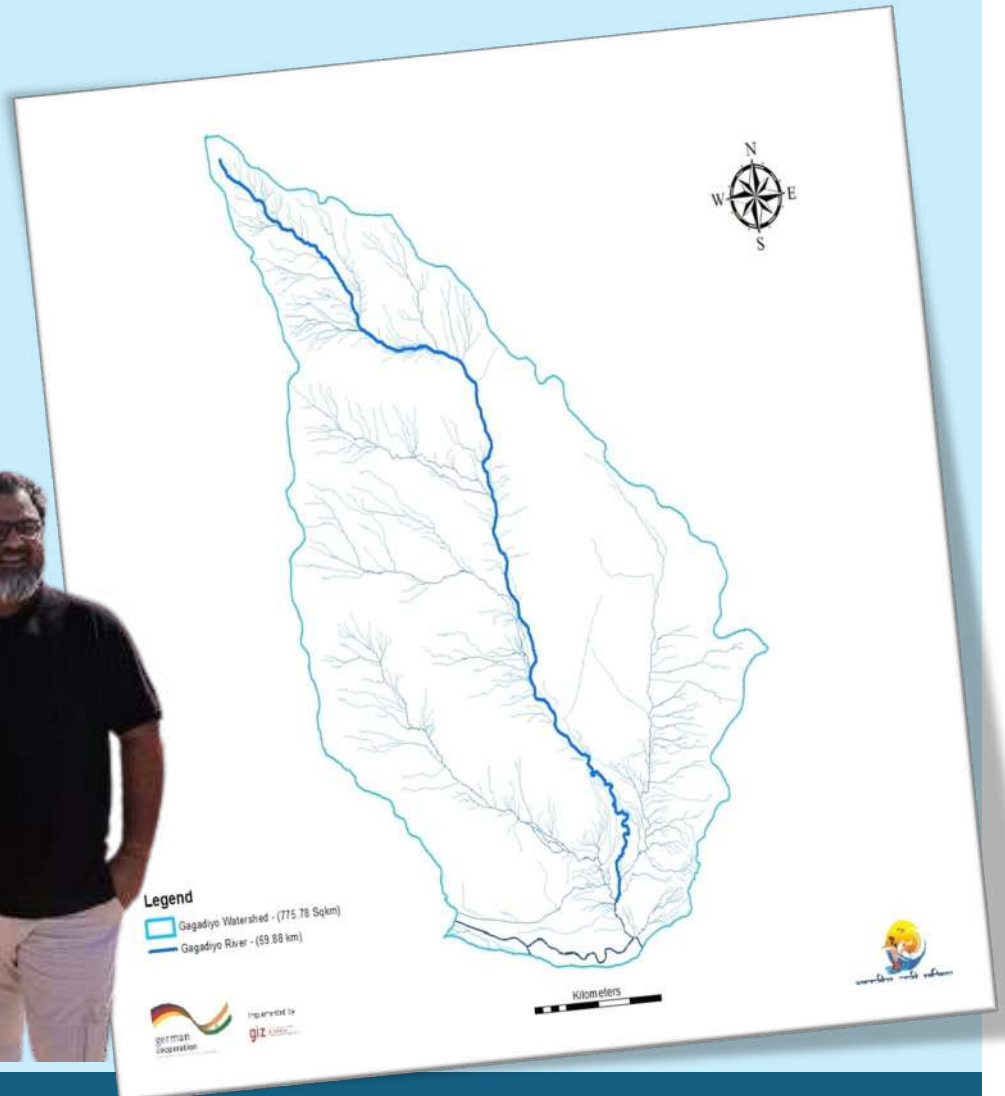




Gagadiyo River Rejuvenation

Amreli District, Gujarat

Report Submitted by:

Bhatiya Nadi Parishad

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My Visit to DHOLAKIA FOUNDATION

(4-6 June, 2024)

A Karmayogi engaged in the work of the God

A slim and healthy body owner, with amazing quickness. His age is evident in the whiteness of his beard and the hair on his head, but even young people are ashamed of the intensity with which he approaches his task. His uncomplicated personality gives him influence over both the elite and the common man. firm in his beliefs. a devoted follower of Krishna, a patriot, and a lover of nature who maintains family unity. Faithful in his objectives, the bearer of wealth and a representation of courage and unity. Named Shri Savji Dholakia, he is a big diamond merchant despite looking like a Fakeer. People love to call him Kaka.

Kaka was born in a small village Dudhala in Amreli district of Gujarat. He came to Surat for business. When he became prosperous, he took a vow to work for the welfare of the village and started doing social work. Although the list of his social works is long, but since 2017 he is completely doing the work of nature i.e. God. He has a full family. Kaka has four brothers. All the four brothers have 8 sons and 6 daughters. There are parents. The whole family is involved in business with the spirit of good and profit, but Kaka also does the work of nature along with business. He has also received the fourth highest award of the Government of India, Padma Shri, for doing the work of God.



It is the greatness of Kaka that he started the work of nature from the same place where he was born, spent his childhood and received his initial education. Kaka has initially taken up the task of reviving the river Gagadiya of his village and the reservoirs there. He does this work through Dholakia Foundation. In the last seven years, many large reservoirs have been revived by him and this work is still going on continuously. Kaka's first big reservoir was built in Dudhala village with the inspiration of the Hon'ble Prime Minister Mr Narendra Modi, which was inaugurated by the Prime Minister himself. As soon as the reservoirs on the river bank were deepened, they filled up with water during the rainy season. After this, a water festival was also celebrated there, in which thousands of people from all over the country participated.

For the last two months, I was in touch with Kaka's nephew Mr. Brijesh Dholakia. Meanwhile, Ms. Charu Mishra from Dholakia Foundation informed me that I am being invited as the chief guest on the occasion

of World Environment Day to be organized by Dholakia Foundation on 5 June. Ms. Jhanvi Patel of Dholakia Foundation was constantly updating me regarding my Surat trip.

I had just finished my address at the Dholakia Foundation event in Surat when there was some commotion and Kaka and his younger brother Mr. Tulsi Dholakia entered the auditorium, walking fast and wearing caps on their heads. I also greeted Kaka and listened to his meaningful and energetic address. On this occasion, Kaka presented me a shawl, which was a proud moment for me.



After the program ended, Kaka showed me his office, told me about his business and had me eat with him. After this, we reached Amreli with Kaka from Surat Airport by a small plane and from there directly to Kaka's Haveli in Dudhala village. Kaka showed us his ancestral house and school in the village. Solar panels are installed in every house in Dudhala village. Kaka's Haveli is situated on the banks of the Gagadiya river, which is at some distance from his ancestral house. We stayed there for some time and then left to see the works on the Gagadiya river. Kaka himself had come on the

charioteer's seat in the Scorpio car and made me sit in the front. Kanak Bhai, who stays in Dudhala and oversees the works of river rejuvenation, was also with us.

We visited about ten big reservoirs and about 30 kilometers of river. Work is being done simultaneously in the entire river. A team of about 200 peoples is engaged in this work. Dozens of machines are working day and night. Some check dams have been built while the work on about two dozen check dams is in its final stages. The river bed has been deepened from 10 to 40 feet depending on the location. Lakhs of plants of different species have been planted along the river, most of which have now become trees. Orchards of mango, berry, fig, guava and peach have been planted in hundreds of acres of land, which have now started bearing fruits. Various arrangements have been made to provide water to all the trees and plants easily.



A big cowshed has been built at one place on the river bank and a Nandishala at another place, in which hundreds of Gir breed cows and bulls are being reared. Kaka kept stopping at every check dam in between and kept watching the work, kept telling me in detail and also kept giving necessary instructions in Gujarati language to the workers working there. Kaka would stop the car and quickly jump here and there on the dam. Such enthusiasm can be seen only in the person who enjoys his work.



A bird park has also been built on an island in the middle of the river. Birds of various foreign species are present in it. I also got the opportunity to feed these birds with Kaka. While we saw herds of lions, animals like Neelgai, wild boar, goat, cow, dog on the river bank, various types of aquatic creatures were seen chirping in the river water. Due to Kaka's efforts for river rejuvenation, Gagadiya river has become the center of complete biodiversity.



When we returned from the river journey in the evening, farmers started coming to the Haveli gradually. Everyone kept greeting Kaka and kept sitting. Kaka introduced me to everyone as well. The river conversation with all the farmers continued till 11.30 pm.

Kaka is working very hard for Gagadiya river rejuvenation. In exchange of the lands of the farmers coming in the way of the river, he has given them many times more land at other places. Kaka goes from Surat to Dudhala every week to see the

work of river rejuvenation. This is his routine for the last seven years.

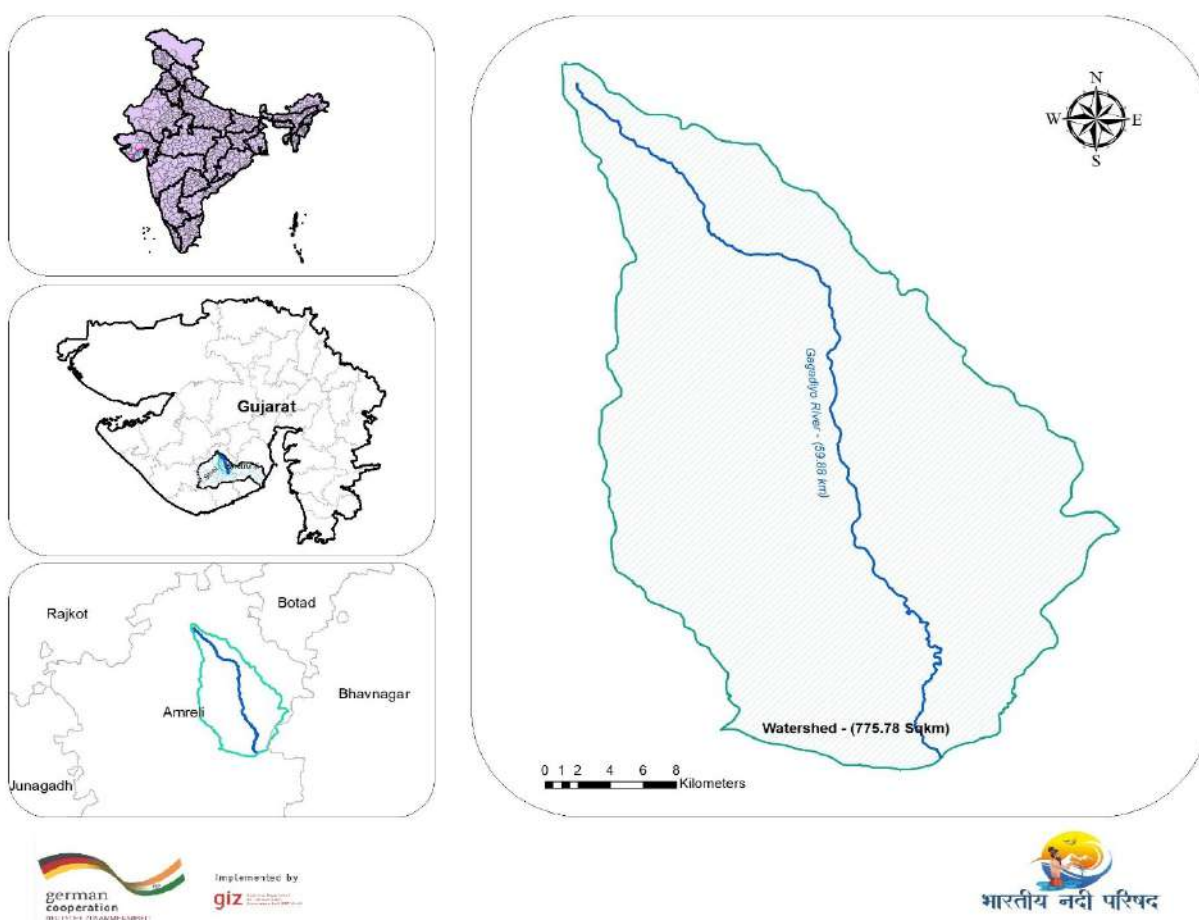
I have seen hundreds of rivers in the country and the world and have closely understood the efforts being made for their revival, but the honesty, dedication, hard work, commitment, sacrifice and clarity with which Kaka is doing the work of reviving the river Gagadiya is unmatched in the country. I can say from my social experience of the last 22 years that in the coming 2 to 5 years, the work of reviving the river Gagadiya will set an ideal example for the country and the world.

India is home to numerous rivers and natural drainage systems that play a critical role in supporting its diverse ecosystems, agricultural practices, and human settlements. However, many rivers in India face severe degradation due to pollution, over-extraction of water, deforestation, and urbanization. The decline in river health has resulted in loss of biodiversity, reduced water quality, and diminished ecological functions. Gagadiya River in Gujarat state is also an example of the same degradation that needs to be rejuvenated.

Gagadiya River: An Introduction

Gagadiya River is referred in the *Gujarat State Gazetteer (amreli) Vol 12 (1972)* in the last line of the First Paragraph of the heading ‘**Drainage: River System**’ under the Chapter ‘**General**’ (Page No. 9) where it is mentioned as ‘There are also other minor rivers and rivulets called **the Gagadiyo**, the Rampar, the Thebi, the Satali, the Dhatarwadi and the Raydi.’ The Gazetteer of the Amreli Prant in Gujarati was First prepared by R. P. Govindbhai Hathibhai Desai of the Baroda State Service and was published in 1920. Besides, the Gazetteer of the Baroda State in English which was prepared by R. B. Govindbhai Hathibhai Desai and Mr. A. B. Clarke and published in 1923 included references to the Amreli district.

Location Map of Gagadiya River

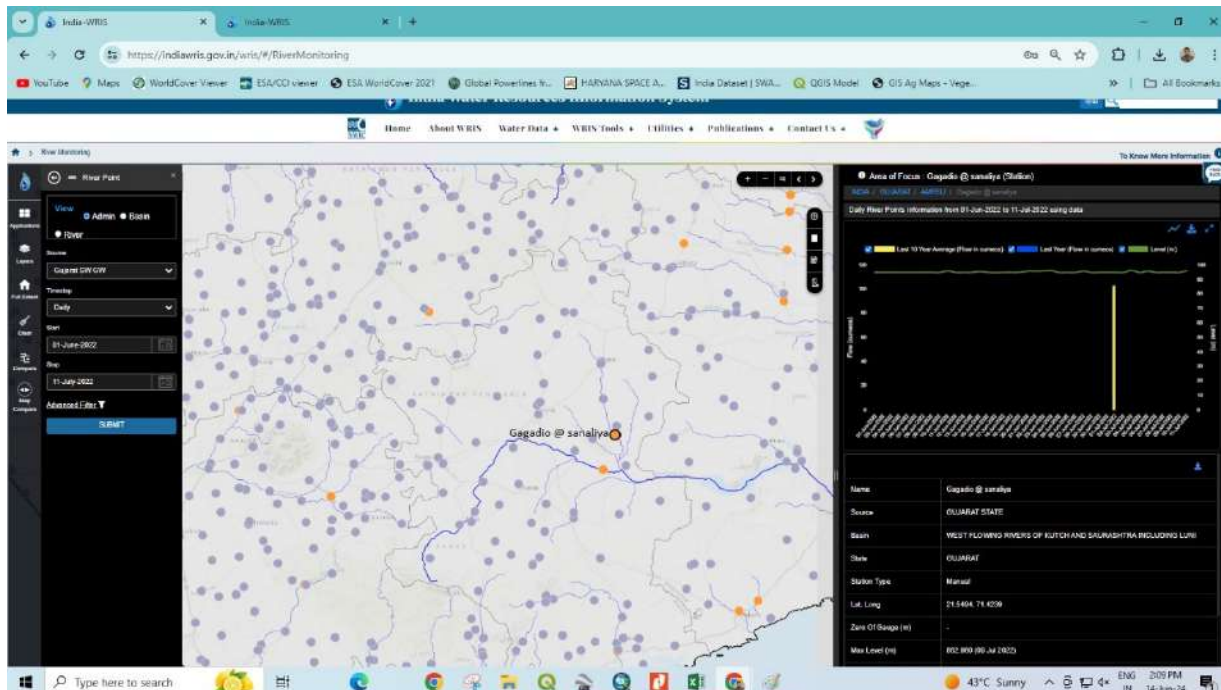


Gagadiya River is situated in Amreli District of Gujarat State in India. It is a part of Shetrungi River Basin with Watershed area of 775 square kilometers. Gagadiya River originates from the coordinates 21.843634, 71.245772 and ends at 21.480574, 71.432629. The total length of Gagadiya river is 59.88 Km.

No.	Layer Name	Area/Length
1	Gagadiyo Watershed Area	775.78 Sqkm
2	Gagadiyo River Length	59.88 km
3	Gagadiyo River Starting Coordinate	(21.843634,71.245772)
4	Gagadiyo River Ending Coordinate	(21.480574,71.432629)

Status of Gagadiyo River:

Max Discharge of Gagadiya River is 206.13 Cumecs and Min Discharge is 30.2 Cumecs. The maximum elevation recorded in Gagadiya river is 862 m and the annual rainfall in the watershed is 428 mm per year. The annual rainfall recorded in the Gagadiya Watershed is 428.6 mm per year.



No.	River Discharge Station Name Gagadio @ Sanaliya	Area/Length
1	Max Level (m)	862.96
2	Max Discharge (cumecs)	206.13
3	Min Discharge (cumecs)	30.2
4	Rainfall in Gagadiyo Watershed	428.6 mm/yr



Constraints and Challenges Related to Gagadiya River

1. **Pollution:** Industrial discharge, agricultural runoff, and untreated sewage contribute significantly to river pollution, affecting water quality and aquatic life.
2. **Over-extraction of Water:** Excessive withdrawal for agriculture and drinking water reduces river flow, affecting ecosystems and leading to the drying up of river stretches.
3. **Deforestation and Land Use Changes:** Deforestation in catchment areas leads to soil erosion, reduced groundwater recharge, and increased sediment load in rivers.
4. **Climate Change:** Altered precipitation patterns and increased frequency of extreme weather events disrupt the natural flow regimes of rivers.
5. **Encroachments:** Urbanization and encroachment on riverbanks reduce the natural drainage capacity and lead to increased flooding.



How to improve Gagadiya river rejuvenation work

The work of Gagadiya river rejuvenation should become a good example of river improvement for the country, so that river workers, organizations and even the government should take inspiration from it and take forward the work of the river. Along with the efforts currently being made for the Gagadiya river, some such efforts are also necessary which can also meet the test of scientific approach and social uniformity. First of all Gagadiya River Council will be formed for the rejuvenation of Gagadiya river.

Structure of Gagadiya River Council:

Gagadiya River Council will be formed by the Pradhans and Secretaries of the villages situated on the banks of the river in all the districts of the river flow area. Out of all these Pradhans, one will be elected as the President, Vice President and Secretary of the Council on the basis of majority. The President and Vice President will be selected from among the Pradhans while the Secretary will be a representative of Dholakia Foundation. By doing so, there will be no problem in working from a technical point of view.

Gagadiya River Council is a group that will take all the social decisions regarding the arrangement of clean and uninterrupted flow of Gagadiya and its tributary rivers. It will be a social organization formed at the level of society. This organization will take care of its river and will also take decisions for its betterment. All decisions will be under Indian rules and regulations and will be taken without any discrimination. The decisions taken by Gagadiya River Council regarding Gagadiya River will have to be informed to Dholakia Foundation because Gagadiya River Council will be a part of Dholakia Foundation which will be provided technical support by Indian River Council and GIZ India.

Village Committee:

The Pradhans of the villages along the river flow will be included in the Gagadiya River Council. It will be the responsibility of the Pradhan of each village to form a committee of ten or more members of such aware people under his chairmanship in his village, which will do the work of river improvement in that part of the village. The village Pradhan will appoint an aware person as the Vice President on the basis of majority. The meeting of the village committee will be held on the 2nd of every month at the place and time decided by the consent of the Pradhan and the Vice President. In the meeting, the old works will be reviewed and the outline of new works will be made. The Vice President will have the task of writing down the agenda of the meeting and the decisions taken during the meeting. A register will be made for this. Although this village committee will also be a part of the Gagadiya River Council, only the village Pradhan and the Vice President will participate in the meetings of the council.

Functioning of Gagadiya River Council:

Gagadiya River Council will meet on the first date of every month at a decided place (place will be a village or mansion on the river bank) and time with the consent of the chairman and secretary. In the meeting, the previous works will be reviewed and plans for future works will be made. The secretary will have the task of writing down the agenda of the meeting and the decisions taken during the meeting. If the cooperation

of the administration is also taken in making plans for the decisions taken in the meeting and implementing those works on the ground, then the work will be completed in a better way.

Rights of Gagadiya River Council to take decisions:

Gagadiya River Council will have the following rights to take decisions-

- *Related to cleaning of the river.*
- *Related to planting trees on the river bank.*
- *Related to revival of ponds on the river bank or in the villages on the river bank.*
- *Related to stopping liquid and solid waste in the river.*
- *Related to public awareness in the villages on the river bank.*

Apart from this, the Gagadiya River Council will be able to take any decision regarding the betterment of the river under the administrative rules and regulations.

Publicity

A WhatsApp group of the members of the Gagadiya River Council will be formed. All the members of the Gagadiya River Council will be included in it. Apart from this, people participating in the river improvement work in the village can also be included with the consent of the Council Chairman. From time to time, news related to the river works and the decisions of the meeting can also be given to the local newspapers with the consent of the Council Chairman. Apart from this, creating a website, creating a Facebook page and creating a Twitter handle etc. can also be used to promote the work being done for the river.

From my experiences, I am presenting the following suggestions to make the work of Gagadiya river rejuvenation permanent and inclusive.

It is a well-known fact that any river flows due to the meeting of groundwater and surface water. In the rainy season, surface water increases the groundwater and after the rains, groundwater provides water to the surface of the river.

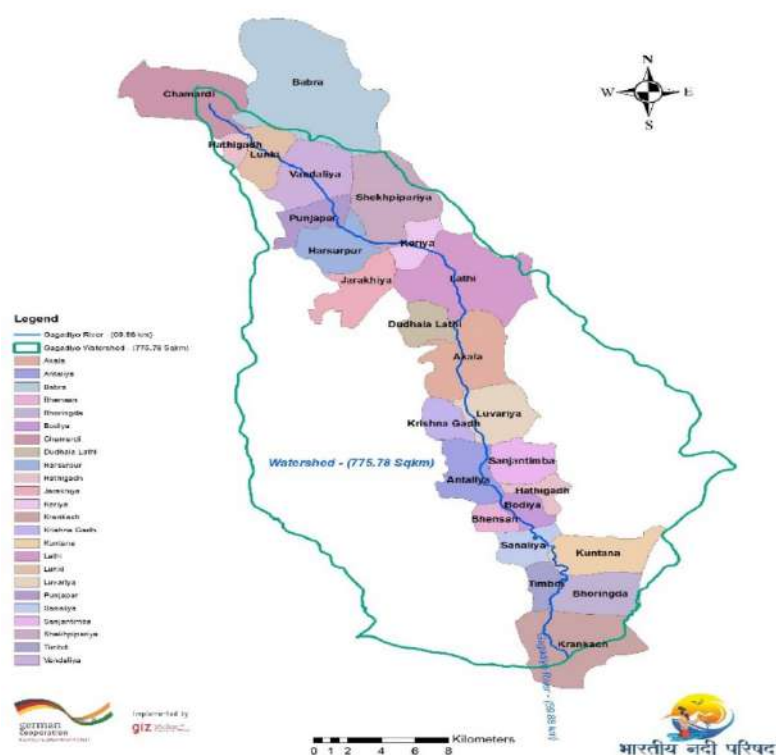
It is also a well-known fact that in proportion to the flow of the river, the same amount of land on both the banks of the river is also the river, which is called the catchment area of the river. This catchment area of the river plays the most important role in keeping the river clean and uninterrupted.

Keeping both the above facts in mind, we should take forward the efforts of river revival. For any river revival, it is advisable to work through a fixed format. The river revival format is enclosed.

Action Research and Development

❖ *A need assessment study of the Gagadiya river should be done.*

Village in 1km Buffer Map



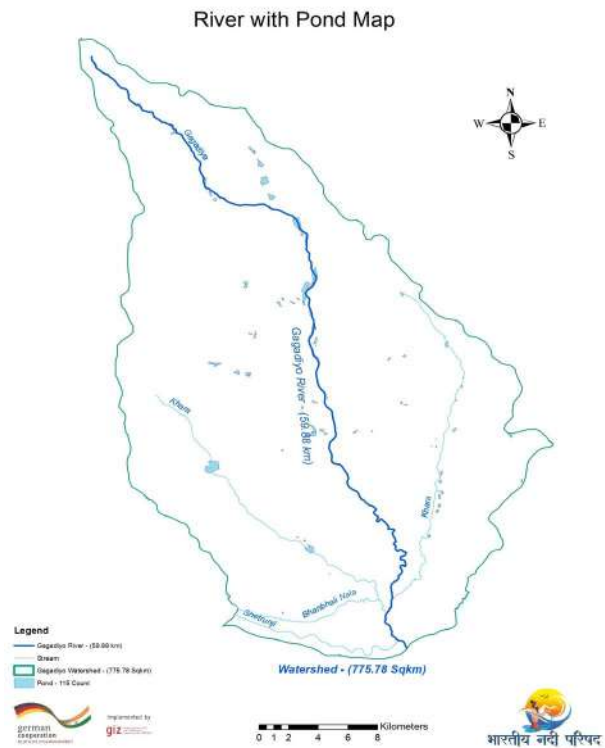
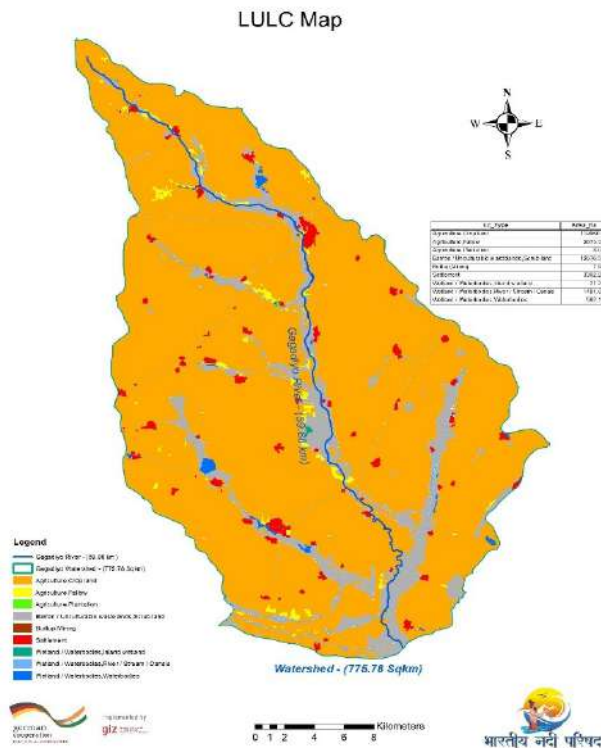
Gagadiya River Council

Gagadiya River Council			
S. No.	Name	Mobile Number	Designation
1	Elected Pradhan Name		President
2	Elected Pradhan Name		Vice President
3	Dholakia Representative		Secretary
4	India River Council Representative		Technical Expert
5	GIZ India Representative		Technical Expert
6	Other Pradhans		Member
7	Other Panchayat Secretaries		Member

❖ Research should be done on the entire biodiversity of the river and it should be documented. Which includes the following research -

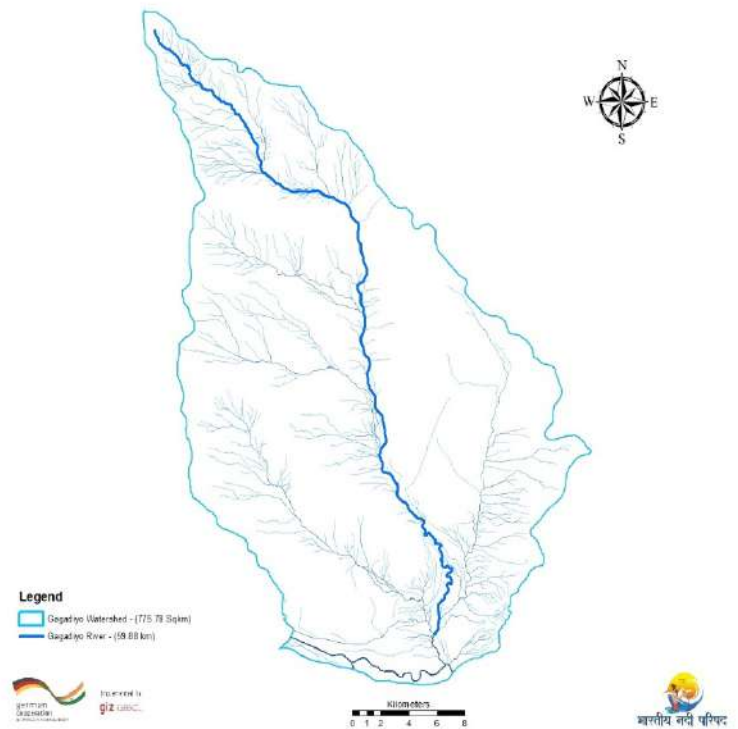
- Gagadiya River and agriculture
- Gagadiya River and water
- Gagadiya River and culture
- Gagadiya River and forest
- Gagadiya River and animals
- Gagadiya River and birds

❖ There are a total 125 ponds/water bodies in the catchment of Gagadiya River which can be reconstructed or rejuvenated. These measures not only reduce runoff but also promote groundwater recharge, which is essential for maintaining base flows in the river.

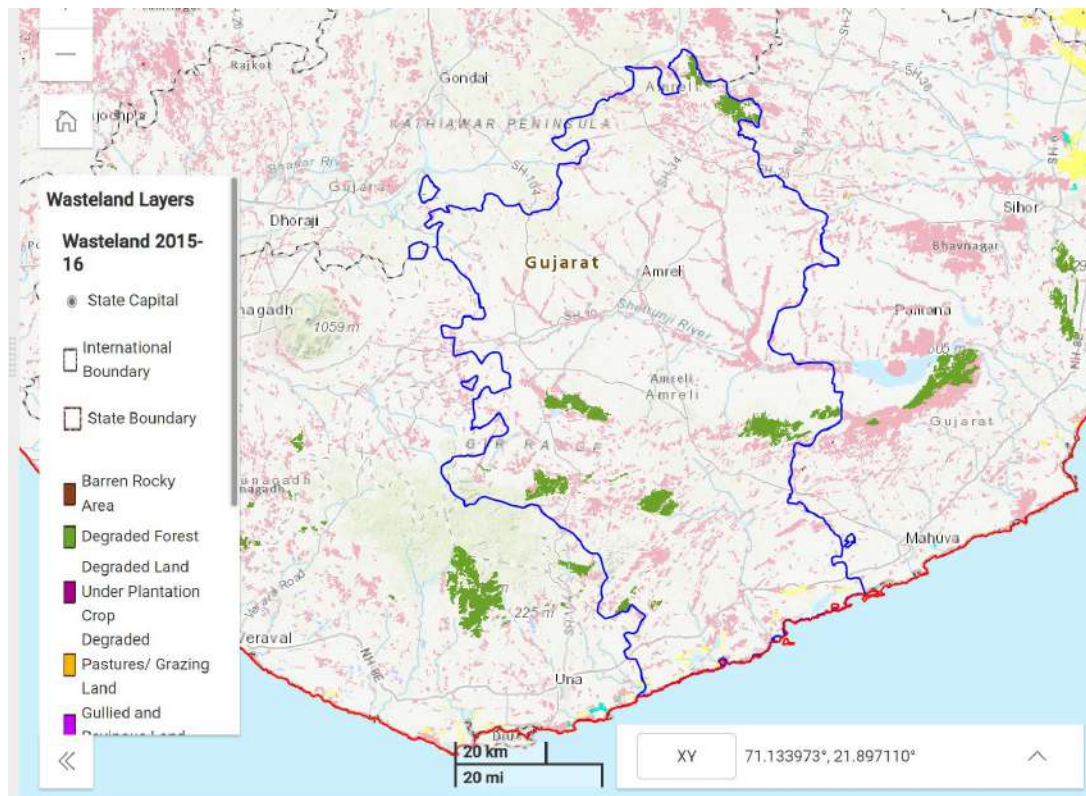


❖ *Drainage Line Treatment of small order drains in the catchment of Gagadiya river need to be done on Priority basis. Drainage line treatment focuses on stabilizing and protecting drainage channels to prevent soil erosion and improve water flow. This can be achieved through the construction of gabions, contour trenches, and vegetative barriers along drainage lines. In the Gagadiyo River context, implementing these measures can help restore the natural drainage system and reduce sedimentation in the river.*

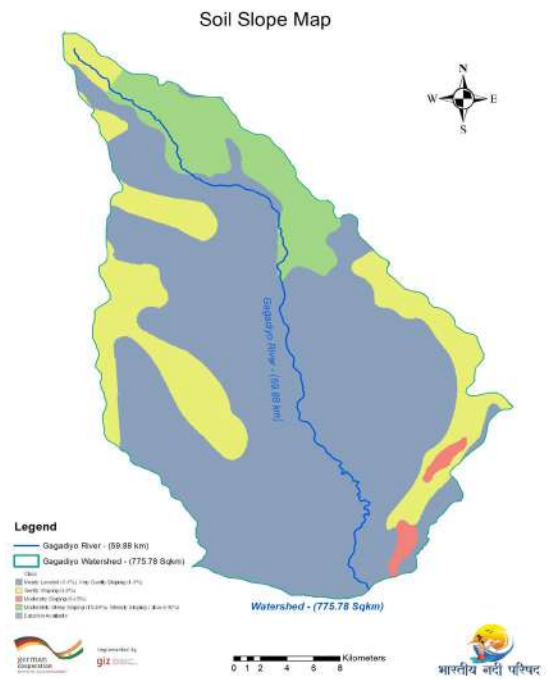
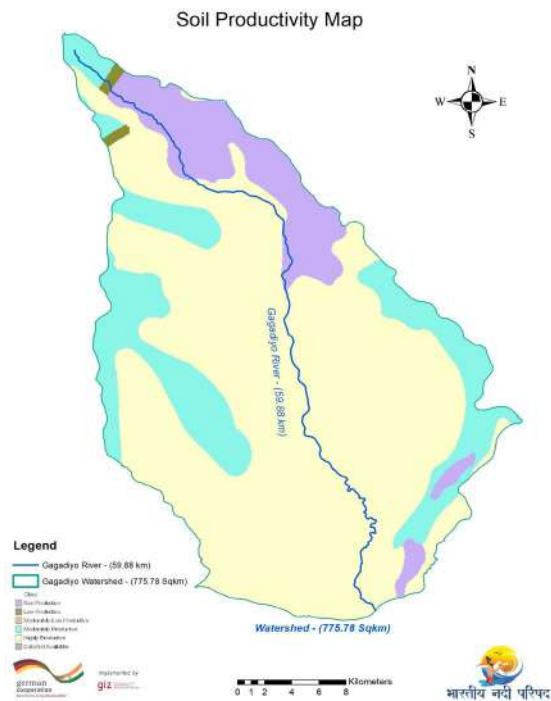
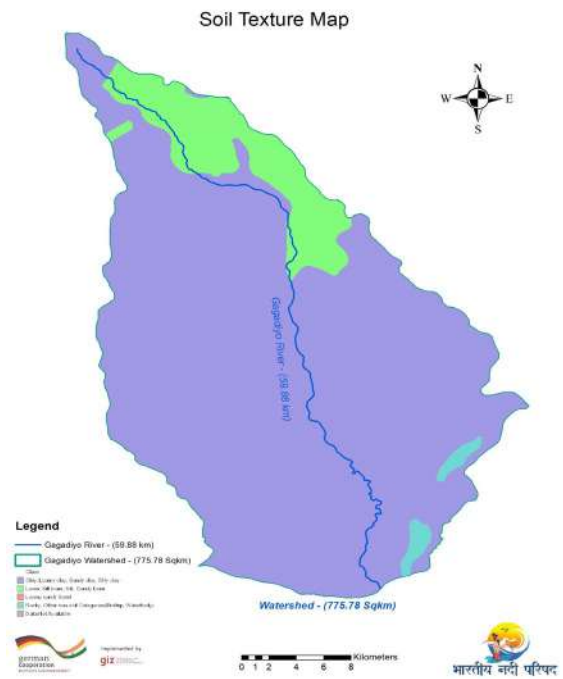
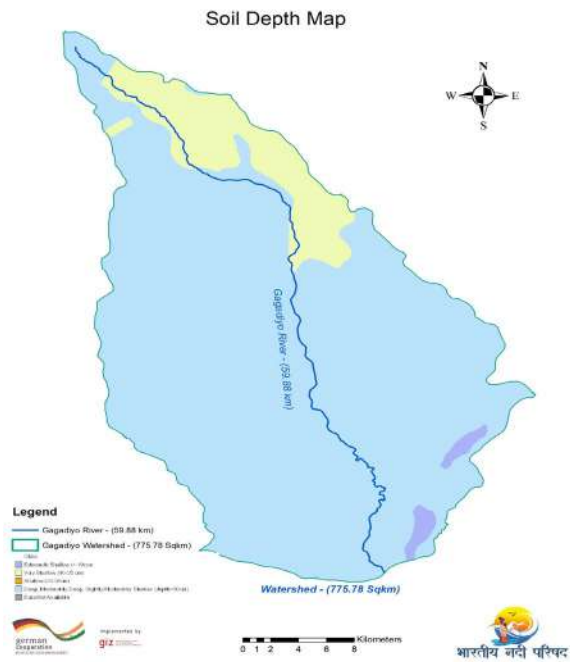
Drainage Map of Gagadiya Watershed



❖ Wastelands



❖ Different types of soil are present in the Gagadiya River. For this, samples of all these types of soil should be tested. It should be published under the name 'Soils of Gagadiya River'.



- ❖ *A complete documentary and a dozen other documentaries showing different aspects of the river should be made.*

in

S. No.	Organization
1	Bhartiya Nadi Parishad (Indian River Council)
2	GIZ India
3	Namami Gange
4	C-Ganga
5	International River Foundation
6	International River Network
7	Waterkeeper Alliance
8	River Management Society
9	International Rivers

S. No.	Awards
1	International River Award
2	Terry Baker Award
3	National Water Award
4	Jal Prahari
5	Champion of Earth
6	Goldman Award
7	Water Digest Award

❖ We suggest you to send the success story of Gagadiya river rejuvenation in the following newspapers and magazines -

S. No.	Newspaper/Magazines
1	Down Two Earth
2	Terra Green
3	Civil Society
4	Asian Water News
5	Panchjanya
6	Organizer
7	DW Hindi
8	Lallantop
9	Jindgi with Richa
10	The Logical Indian
11	Better India
12	Good News India

❖ *Apart from this, we present the following suggestions to you for furthering the work related to water conservation and river rejuvenation in the state of Gujarat.*

❖ *Prepare a draft of Gujarat's river policy and make Gujarat's river policy on its basis.*

❖ *Prepare an atlas of Gujarat's rivers.*

❖ *Document the historical lakes of Gujarat.*

❖ *Form a pond development authority in Gujarat.*

❖ *We suggest you establish a 'River Research Center' on the banks of the Gagadiya River. In this center, where the river and its rejuvenation will be taught, it will also be taught practically. Here, courses related to river rejuvenation will be conducted from one week to one month. We are also sending you a proposal to establish this center.*

❖ *Herbal gardens should also be developed on the banks of the Gagadiya river.*

- ❖ All the plants planted along the river should be Geo-tagged.
 - ❖ A river tiling route should be made along the river. On this route, boards should be installed at special places giving information about the place, village, dam, reservoir and trees and plants etc.
 - ❖ Fixed binoculars should be installed at special places on the river tiling route. Standing there, one can see the view of the river till a distance. This type of experiment has been done on the Genesee River in America's Letchworth State Park, which is working successfully.
 - ❖ Arrangements for sheds, benches and water should also be made at some distance for sitting on the river tiling route.
 - ❖ The dams built on the river, the forests built, herbal gardens, reservoirs, parks or other things should be named after the village, some brave soldier or famous personality according to the local society.
 - ❖ For this we are sending you some documents, which will help you in taking forward the work of Gagadiya river revival, pond conservation and river and water in Gujarat.
1. River revival format.
 2. Pond revival format.
 3. Draft of river policy of Uttar Pradesh.
 4. Method of formation of Gagadiya River Council and its format.

Conclusion

The Gagadiyo River in the Shetrungi sub-basin holds strategic importance for the Saurashtra region of Gujarat. Addressing the challenges facing this river through comprehensive eco-restoration efforts can significantly improve its health and sustainability. By focusing on catchment area treatment, drainage line treatment, run-off management, and maintaining ecological flow, it is possible to enhance the river's ecosystem services and ensure it continues to support the region's agricultural, ecological, and social needs.

ANNEXURE:1

LIST OF VILLAGES ON THE BANK OF GAGADIYA RIVER

S.No	Village	Subdistric	District	State	LGD_Code11	RuraUrban
1	Hathigadh	Babra	Amreli	Gujarat	515476	Rural
2	Chamardi	Babra	Amreli	Gujarat	515466	Rural
3	Lunki	Babra	Amreli	Gujarat	515477	Rural
4	Vandaliya	Babra	Amreli	Gujarat	515478	Rural
5	Shekhipariya	Lathi	Amreli	Gujarat	515487	Rural
6	Punjapar	Lathi	Amreli	Gujarat	515488	Rural
7	Harsurpur	Lathi	Amreli	Gujarat	515489	Rural
8	Keriya	Lathi	Amreli	Gujarat	515490	Rural
9	Jaraxhiya	Lathi	Amreli	Gujarat	515502	Rural
10	Dudhala Lathi	Lathi	Amreli	Gujarat	515504	Rural
11	Akala	Lathi	Amreli	Gujarat	515515	Rural
12	Luvariya	Lathi	Amreli	Gujarat	515521	Rural
13	Krishna Gadh	Lathi	Amreli	Gujarat	515520	Rural
14	Sanjantimba	Lilia	Amreli	Gujarat	515536	Rural
15	Antaliya	Lilia	Amreli	Gujarat	515535	Rural
16	Hathigadh	Lilia	Amreli	Gujarat	515543	Rural
17	Bodiya	Lilia	Amreli	Gujarat	515550	Rural
18	Bhensan	Lilia	Amreli	Gujarat	515549	Rural
19	Sanaliya	Lilia	Amreli	Gujarat	515555	Rural
20	Kuntana	Lilia	Amreli	Gujarat	515554	Rural
21	Timbdi	Lilia	Amreli	Gujarat	515556	Rural
22	Bhoringda	Lilia	Amreli	Gujarat	515564	Rural
23	Krankach	Lilia	Amreli	Gujarat	515565	Rural
24	Babra	Babra	Amreli	Gujarat	253147	Urban
25	Lathi	Lathi	Amreli	Gujarat	251155	Urban

ANNEXURE:2

Yearly rainfall(in mm) trends for Shetranjuli and other east flowing rivers from 01-Jan-2013 to 31-Dec-2023

Dates	NORMAL (mm)	ACTUAL (mm)
2013	621.12	975.58
2014	621.12	563.04
2015	621.12	631.49
2016	621.12	679.58
2017	621.12	639.2
2018	621.12	514.44
2019	621.12	924.04
2020	621.12	930.25
2021	621.12	784.33
2022	621.12	671.28
2023	621.12	770.6

ANNEXURE:3

Wastelands in Amreli District forming the catechment area for Gagdiyo River

S.No	Class	Area (Ha)
1	Barren rocky area	7.805594
2	Degraded forest	17309.36
3	Degraded land under plantation	0
4	Degraded pastures grazing land	0
5	Gullied and ravinous land	0
6	Mining industrial wastelands	53.83002
7	Salt affected area	1268.803
8	Sandy area	10.10226
9	Scrub land	60529.41
10	Shifting cultivation	0
11	Snow covered glacial area	0
12	Waterlogged area	37.48489
	Total	79216.79